Triadic Casimir — Fourier Atlas v0.4.1 GPU Framework Report

SST Workbench / Unregistered instance

October 19, 2026

1 Instance identity

A065-v0.4.2 numerical GPU Arc A770 Euler qualification instance.

2 Research question

Does a preregistered triadic clock/stress geometry pilot satisfy its analytic controls while independent physical coupling gates remain unresolved?

3 Null and alternative hypotheses

No independently qualified source-specific physical two-surface response is established by the present geometry and toy models.

4 Explicit assumptions

Time-dependent regularized filament RK4 is exploratory and is not full Euler closure.

5 Definitions

u material velocity; omega=curl u; S symmetric strain.

6 Analytical equations

The official AUTO_SCIENCE_CONTRACT fragment records frozen equations.

Machine-readable science contract

Research question Does a preregistered triadic clock/stress geometry pilot satisfy its analytic controls while independent physical coupling gates remain unresolved?

Objective Independently verify three dataset seals, analytic Euler/gauge/sheet controls and one sourced KnotPlot trefoil finite-core geometric response without importing target force.

H_0 No independently qualified source-specific physical two-surface response is established by the present geometry and toy models.

H_1 Independently qualified PKLSA Euler motion, boundary conditions and microscopic link coupling produce held-out physical predictions.

Registered equations

E1: Incompressible Euler kinematics

$$\nabla \cdot \mathbf{u} = 0, \quad \boldsymbol{\omega} = \nabla \times \mathbf{u} \tag{1}$$

Dimensional check: divergence and vorticity both inverse seconds

E2: Pressure Poisson source

$$\nabla^2 p = \rho_M(\frac{1}{2}|\omega|^2 - ||S||_F^2) \tag{2}$$

Dimensional check: kg/(mˆ3 sˆ2) on each side

E3: Regularized geometric velocity diagnostic

$$\mathbf{u}(\mathbf{x}) = \frac{1}{4\pi}\sum_j \frac{\Delta\mathbf{X}_j \times (\mathbf{x} - \mathbf{X}_{j+1/2})}{(|\mathbf{x} - \mathbf{X}_{j+1/2}|^2 + \varepsilon^2)^{3/2}} \tag{3}$$

Dimensional check: dimensionless Gamma=1 geometry diagnostic

E4: Profile-dependent cross-sectional enstrophy inverse

$$Z_G = \Gamma^2/(2\pi a^2), \quad Z_R = \Gamma^2/(\pi a^2) \tag{4}$$

Dimensional check: Gammaˆ2/Z = Lˆ2 for SI cross-section

E5: Rosenhead regularization swirl and steady radial pressure balance

$$v_{RM} = \Gamma r/(2\pi(r^2 + \varepsilon^2)), \quad \partial_r p = \rho v_\theta^2/r \tag{5}$$

Dimensional check: pressure gradient M/(Lˆ2 Tˆ2)

E6: 3D Gaussian-smoothed closed vortex current

$$\boldsymbol{\omega}(\mathbf{x}) = \Gamma \oint_C e^{-|\mathbf{x}-\mathbf{X}|^2/a^2}\, d\mathbf{X}/(\pi^{3/2}a^3) \tag{6}$$

Dimensional check: omega 1/T if Gamma Lˆ2/T

E7: Local cross-section core inference under Gaussian profile

$$\Gamma_\perp = \int_\Sigma \omega_\parallel dA, \quad Z_\perp = \int_\Sigma \omega_\parallel^2 dA, \quad a_G = |\Gamma_\perp|/\sqrt{2\pi Z_\perp} \tag{7}$$

Dimensional check: Gammaˆ2 / Z = Lˆ2

E8: Periodic velocity and Poisson pressure diagnostics

$$\hat{\mathbf{u}} = i\mathbf{k} \times \hat{\boldsymbol{\omega}}/|\mathbf{k}|^2, \quad \nabla^2 p = -\rho \partial_i u_j \partial_j u_i \tag{8}$$

Dimensional check: periodic numerical diagnostics only

Registered code-to-mathematics steps

1. **S1** [G1; equations E1]: Verify all three source seals before any scientific result is evaluated
2. **S2** [G2; equations E2, E3]: Evaluate analytic Euler controls and preregistered trefoil geometry resolution scan
3. **S3** [G3; equations E3]: Withhold physical Casimir claims without independent providers, wall dynamics, and atom-link normalization
4. **S10** [G15; equations E3]: Verify direct parametric XYZ N600 to N4800 hashes and nesting
5. **S11** [G16; equations E3]: Independent fixed-core 1 percent spatial convergence at baseline epsilon=0.12 across BOTH historical trefoils

6. **S12** [G17; equations E3]: Report eps 0.06 0.12 0.24 numerical core sensitivity without physical force normalization
7. **S13** [G18; equations E3]: Check C++17 1200 and 4800 time step refinement and Python N150 parity
8. **S7** [G12; equations E3]: Audit historical Fourier and torus XYZ hash and analytic geometry
9. **S8** [G13; equations E3]: Compare reduced vortex filament RK4 spatial/time resolution on independent geometric seeds
10. **S9** [G14; equations E3]: Report fixed core regularization sensitivity without physical force interpretation
11. **S4** [G10; equations E3]: Verify fixed-core 150/300/600 convergence, separately retaining original G2 failures
12. **S6** [G3; equations E1, E2, E3]: Evaluate G3 only after precommitted G1+G10 and temporal/analytic guard; retain G2=FAIL and physical UNRESOLVED
13. **S5** [G11; equations E3]: Verify exact Fremlin raw hash and native C++ time-series refinement; never infer real Euler solution
14. **S19** [G19; equations E3]: Regenerate 2015 parametric trefoils inside falsifier, hash-check vs pinned XYZ and original generator
15. **S20** [G20; equations E3]: Measure mean segment length divided by fixed regularization epsilon at each resolution without fitting
16. **S21** [G21; equations E3]: Test native C++ RK4 fine-mesh spatial convergence for regenerated trefoils at epsilon=0.06
17. **S22** [G22; equations E4, E5]: Verify precommitted finite-core analytic source
18. **S23** [G23; equations E4, E5]: Invert independently specified Gamma and Z
19. **S24** [G24; equations E4, E5]: Hold out vortex and pressure-gradient comparisons, no cutoff refit
20. **S25** [G25; equations E4, E5]: Test mapping nonuniqueness
21. **S26** [G26; equations E4, E5]: Verify native FP64 kernel parity
22. **S27** [G27; equations E4, E5]: Keep physical core unqualified without external vorticity cross section
23. **S28** [G28; equations E6, E7, E8]: Generate source-classified 3D surrogate and check solenoidal projection
24. **S29** [G29; equations E6, E7, E8]: Reject contaminated transverse sections
25. **S30** [G30; equations E6, E7, E8]: Invert Gamma/Z without velocity pressure fitting
26. **S31** [G31; equations E6, E7, E8]: Evaluate withheld dependent numerical u and pressure without retrospective calibration
27. **S32** [G32; equations E6, E7, E8]: Require upstream native Euler snapshot and independently computed physical holdout
28. **S33** [G33; equations E3]: Verify unmodified external A003 adapter implementation
29. **S34** [G34; equations E3]: Validate full A003 atlas source manifest and ALL 384 original XYZ records
30. **S35** [G35; equations E6, E7, E8]: Generate explicitly synthetic volumetric vortex from separately verified A003 Fourier trefoil and diagnose withheld velocity/pressure
31. **S36** [G36; equations E1, E2]: Do not promote geometric surrogate to physical source-native 3D Euler without independently qualified time-dependent field
32. **S43** [G43; equations E8]: Validate actual Arc A770 Level Zero device, no iGPU or CPU substitution
33. **S44** [G44; equations E8]: Euler RK4 FFT Taylor-Green FP32 GPU vs FP64 CPU parity
34. **S45** [G45; equations E8]: Compare 48 64 96 A003 Gaussian-initialized trajectories GPU to CPU FP64
35. **S46** [G46; equations E8]: Compare periodic numerical pressure GPU and CPU, no external holdout
36. **S47** [G47; equations E8]: Require full ladder and source provenance on device-qualified GPU run

37. **S48** [G48; equations E8]: Withhold physical PKLSA core and independent pressure claims
38. **S37** [G37; equations E6]: Authenticate same two A003 geometries for all three grids
39. **S38** [G38; equations E6, E7]: Volume convergence with fixed Gaussian core, no fit
40. **S39** [G39; equations E7]: Independent Fourier pseudospectral Euler RK4 benchmark, Taylor-Green
41. **S40** [G40; equations E7]: Evolve A003-initialized numerical Euler fields for 48/64/96, test energy and divergence
42. **S41** [G41; equations E7]: Keep PKLSA source-native Euler and independent physical pressure unresolved absent provider
43. **S42** [G42; equations E8]: Compare no-fit numerical Poisson pressure over volumes, distinct from physical holdout

7 Scientific algorithm

Verify three seals; compute controls; evaluate trefoil diagnostic.

8 Dataset provenance

D01 D02 D03, KnotPlot N300, external Fremlin and separate 2015 historical Fourier and torus geometric seeds. No historical geometry is physical PKLSA evidence.

9 Independent gates

G12-G14 remain historical tests; G15-G18 separately qualify direct N600-N4800 geometries and reduced ODE, independent of physical G3. No physical conclusions.

10 Numerical controls

Dimensionless finite-segment Biot-Savart and finite-difference derivatives.

11 Backend authority

CPU profile enables strict C++ gate; G3 may be evaluated but physical validation remains unresolved.

12 Blindness and integrity

All three dataset SHA256 seals checked; reveal comparator not used in blind kernel.

13 Statistics

Geometry subsamples are not independent upstream providers.

14 Result status

Read dynamic AUTO_RESULTS fragment; no target-matched force.

Automatic run provenance

Field	Recorded value
Run ID	20261019T211808791055Z-ac5417c5
Run phase	BLIND
Protocol bundle SHA-256 (byte exact)	7015a59c4533d371f2fb3261da439b09e70e428e3244158a0147d772affac284
Portable protocol diagnostic SHA-256	05ac87d8e9d93b609943d035e103cf9f44c63bcde3de35dfb9c86426cd0e977d
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Science evidence SHA-256	de23a3e3b812430e18bfa3eccd237f621cfec79491eac69d2cc25a747e1ec322
Final output manifest	See OUTPUT_MANIFEST.json (finalized after PDF and report metadata)

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Is the frozen protocol intact and is the runtime self-consistent?	Frozen protocol verified by canonical framework.
G1	PASS	Are all preregistered sources present, identified and provenance-qualified?	Source bytes and single shared provider manifest verified; dynamic independence not implied.
G2	FAIL	Do reference numerics satisfy analytic/sanity and dimensional checks?	75/150/300 fixed-core validation. Retain historical FAIL; no tolerance changes.
G10	PASS	Does refined KnotPlot spatial RK4 filament discretization pass 150/300 and 300/600 with fixed epsilon?	Refined filament ODE only; 600 points interpolated from 300. Historic G2 FAIL retained.
G11	FAIL	Does authentic Fremlin C++17 provider satisfy source, spatial and time checks (simulation only)?	C++17 Fremlin numerical provider only, not an independent physical Euler model.
G12	PASS	Are historical Fourier and torus XYZ inputs hash-valid and analytically consistent?	Historical initial geometries verified, not dynamically relaxed PKLSA.
G13	FAIL	Do both historical geometries satisfy preregistered reduced filament ODE resolution and C++ Python parity?	G13 reduced filament ODE resolution only; any failure retained.
G14	PASS	Were epsilon sensitivity diagnostics recorded without fitting physical claims?	Only deterministic epsilon scan completion; not a physical closure.
G15	PASS	Are direct-analytic historical 600/1200/2400/4800 XYZ hash and nesting controls valid?	All direct parametric coordinate hashes and nesting verified; geometry only.
G16	PASS	Do BOTH historical trefoils meet 1 percent spatial convergence at baseline epsilon 0.12, 600 to 4800?	Fixed 1 percent baseline discretization threshold (not physical Euler).
G17	PASS	Are all three precommitted finite-core epsilon scans and complete 4800 point diagnostics recorded?	Epsilon scan completion only; individual failures retained, no physical core fit.
G18	PASS	Does native C++17 RK4 meet temporal convergence at 1200/4800 and Python spot parity?	Native time-convergence and limited Python parity; reduced ODE, not Euler PDE.
G19	PASS	Does the on-run analytic trefoil generator reproduce sealed XYZ and its arclength parameters?	Runtime 2015 analytical generation agrees with pinned XYZ; numerical initial data only.
G20	PASS	Are N-dependent segment-to-epsilon ratios computed and declared without fitting a physical core?	Resolution relative to numerical smoothing parameter measured, never optimized.

G21	PASS	Do fresh formula-generated N1200/ 2400/4800 trefoils converge at fixed epsilon 0.06 in native RK4?	Freshly generated N1200-N4800 geometry, fixed epsilon=.06, native C++ RK4; 600/1200 remains diagnostic.
G22	PASS	Do the synthetic finite-vorticity source invariants, preregistration and immutable legacy seals verify?	Legacy seals and synthetic analytic core inputs verified; not actual PKLSA.
G23	PASS	Can profile-conditional finite core width a be independently recovered from Gamma and cross-sectional enstrophy?	Recover profile-conditional a from independent Gamma,Z ONLY; synthetic positive/negative controls.
G24	FAIL	Does one slope-matched Rosenhead epsilon predict held-out swirl AND pressure-gradient fields within 5 percent?	FAIL is a scientific rejection of one fixed slope-matched Rosenhead cutoff reproducing the full finite-core profile; no retrofit.
G25	PASS	Do distinct legitimate kernel-equivalence mappings imply materially distinct epsilon values (nonuniqueness diagnostic)?	Multiple kernel matching definitions disagree; no universal epsilon inferred.
G26	PASS	Does an actually compiled C++17 FP64 radial-kernel path match Python, kernel-only?	Actual C++17 cross-sectional analytic kernel parity, not a vortex PDE solver.
G27	UNRESOLVED	Is a measured, independent full-PKLSA transverse vorticity core and separate held-out force/velocity provider qualified?	No authenticated real PKLSA transverse vorticity cross section or independent physical velocity/ pressure provider. No SST r_c, Euler, Casimir or gravity claim.
G28	PASS	Is the generated 3D divergence-free vorticity and Poisson solver numerically consistent (synthetic only)?	Geometry-derived synthetic 3D source only; failed contaminated planes and heldout model predictions retained. NO native PKLSA Euler certification.
G29	FAIL	Are ALL local transverse stations free of cancelling flux and neighbour-strand contamination?	Geometry-derived synthetic 3D source only; failed contaminated planes and heldout model predictions retained. NO native PKLSA Euler certification.
G30	PASS	Can Gamma and Z on admissible stations determine a Gaussian-conditional core width without u/p fitting?	Geometry-derived synthetic 3D source only; failed contaminated planes and heldout model predictions retained. NO native PKLSA Euler certification.
G31	FAIL	Does no-fit Gaussian core prediction reproduce withheld dependent numerical velocity/pressure at 5 percent?	Geometry-derived synthetic 3D source only; failed contaminated planes and heldout model predictions retained. NO native PKLSA Euler certification.
G32	UNRESOLVED	Is a 3D dynamic PKLSA source plus INDEPENDENT u/p holdout fully proven?	True PKLSA finite-core time evolution and independent velocity/ pressure targets unavailable.
G33	PASS	Does the external A003 v0.2.3 adapter match its pinned, unmodified release hashes?	Original A003 source and release metadata SHA256 verified; no code vendoring
G34	PASS	Does the original A003 v0.1.2 atlas pass adapter v0.2.3 full geometry and hash audit?	Real A003 adapter validated all original atlas records; geometry only
G35	FAIL	What is the Gaussian surrogate 3D velocity pressure and transverse core behavior of a verified A003 Fourier trefoil?	Explicit Gaussian synthetic core diagnostics on original A003 3_1 seeds; independent physical targets NOT supplied
G36	UNRESOLVED	Is there independent source-native 3D Euler PKLSA time evolution and held-out physical pressure?	No independent time-evolved source-native 3D Euler vorticity AND physically independent pressure/velocity holdout. A003 geometric atlas cannot supply them.

G37	PASS	Are both original A003 geometry files and exact fixed N1200/stride6 sources authenticated for the 48/64/96 experiment?	Frozen v0.4.0 numerical controls; PASS cannot assert PKLSA physical core.
G38	PASS	Do separately regenerated 48^3,64^3,96^3 Gaussian-initialization u,omega,p fields converge at fixed core and box?	Frozen v0.4.0 numerical controls; PASS cannot assert PKLSA physical core.
G39	PASS	Does a separate, dealiased 3D Euler/RK4 FFT solver pass Taylor-Green time-convergence/divergence/energy controls?	Frozen v0.4.0 numerical controls; PASS cannot assert PKLSA physical core.
G40	PASS	Does genuine time-evolved Euler numerics from authenticated A003-initialized geometry preserve divergence and energy at all three grids?	Frozen v0.4.0 numerical controls; PASS cannot assert PKLSA physical core.
G41	UNRESOLVED	Is there INDEPENDENT source-native PKLSA Euler IVP and separately independently observed physical velocity/pressure holdout?	Physically independent PKLSA Euler and pressure holdout NOT provided; A003 provides static geometry; IVP started from assumed Gaussian vorticity.
G42	PASS	Are withheld PRESSURE fields grid-convergent for numerical surrogate and Euler IVP with NO physical pressure validation claims?	Frozen v0.4.0 numerical controls; PASS cannot assert PKLSA physical core.
G43	PASS	Was the real Intel Arc A770 Level Zero device explicitly selected with no CPU fallback?	Explicit Arc A770 Level Zero GPU queue; no fallback
G44	PASS	Does the GPU FFT Euler RK4 Taylor Green control match FP64 with pinned thresholds?	Same operator/initial data GPU FP32 vs independent NumPy FP64 Taylor-Green (not a PKLSA physical test)
G45	PASS	Do all A003 48 64 96 GPU Euler velocity and vorticity snapshots match CPU FP64?	A003 trajectories GPU FP32 vs NumPy FP64 at frozen 48/64/96
G46	PASS	Do the numerical Euler pressure fields match independent CPU FP64 across all A003 grids?	Periodically reconstructed Euler pressure GPU FP32 vs CPU FP64; no independent physical pressure measurement
G47	PASS	Is the A003 GPU campaign complete with device and upstream geometry provenance?	Full three-resolution dual-backend result/manifest with explicit A770 identity; source-hash fingerprint required
G48	UNRESOLVED	Is the PKLSA initial finite core and independent physical pressure holdout qualified?	Gaussian initialization from A003 geometry and dependent numerical pressure cannot qualify native PKLSA physics
G5	PASS	Does actual C++17 FP64 filament velocity match Python at 75/150/300? Kernel-only evidence, not full solver certification.	Actual C++17 kernel executed. Full scientific/backend certification remains open.
G3	UNRESOLVED	Does target-free discovery reject the preregistered null under the registered statistic?	G3 evaluated after refined numerical qualification; independent physical force, complete Euler dynamics and microscopic material response remain unestablished.
G4	UNRESOLVED	Does an independent held-out/provenance-family confirmation reproduce the discovery result?	No independent physical heldout; diagnostic attempted without overriding G3.
G6	NOT_APPLICABLE	If enabled, does GPU screening pass CPU to GPU parity without being promoted beyond its registered authority?	Disabled for CPU-only profile
G7	NOT_APPLICABLE	Does the result replicate across preregistered source families/encodings without pseudo-replication?	Disabled for CPU-only profile

G8	UNRESOLVED	Are physical PKLSA Euler, material, surface and independent-provider closure requirements met?	No material surface, independent full Euler or PKLSA dynamics. Diagnostic-only.
G9	DEFERRED	After all blind gates, does reveal verification match the preregistered commitment and permit target comparison?	Reveal is a separate, commitment-verified phase and is not executed during blind runs.

Scientific qualification (separate from measured gate outcomes)

Gate	Evidence qualification	Blocking prerequisites
G0	QUALIFIED	none
G1	QUALIFIED	none
G2	QUALIFIED	none
G10	QUALIFIED	none
G11	QUALIFIED	none
G12	QUALIFIED	none
G13	QUALIFIED	none
G14	QUALIFIED	none
G15	QUALIFIED	none
G16	QUALIFIED	none
G17	QUALIFIED	none
G18	QUALIFIED	none
G19	QUALIFIED	none
G20	QUALIFIED	none
G21	QUALIFIED	none
G22	QUALIFIED	none
G23	QUALIFIED	none
G24	QUALIFIED	none
G25	QUALIFIED	none
G26	QUALIFIED	none
G27	UNRESOLVED	none
G28	QUALIFIED	none
G29	QUALIFIED	none
G30	QUALIFIED	none
G31	QUALIFIED	none
G32	UNRESOLVED	none
G33	QUALIFIED	none
G34	QUALIFIED	none
G35	QUALIFIED	none
G36	UNRESOLVED	none
G37	QUALIFIED	none
G38	QUALIFIED	none
G39	QUALIFIED	none
G40	QUALIFIED	none
G41	UNRESOLVED	none
G42	QUALIFIED	none
G43	QUALIFIED	none
G44	QUALIFIED	none
G45	QUALIFIED	none
G46	QUALIFIED	none
G47	QUALIFIED	none
G48	UNRESOLVED	none
G5	QUALIFIED	none
G3	UNRESOLVED	none
G4	DIAGNOSTIC_ONLY	G3
G6	NOT_APPLICABLE	none
G7	NOT_APPLICABLE	none
G8	DIAGNOSTIC_ONLY	G3
G9	DEFERRED	G8

Backend manifest


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geometry only",
        "status": "PASS"
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          "source_class": "SYNTHETIC_DIAGNOSTIC_ONLY"
        },
        "reason": "Explicit Gaussian synthetic core diagnostics on original
A003 3_1 seeds; independent physical targets NOT supplied",
        "status": "FAIL"
      },
      "G36": {
        "execution_status": "EXECUTED",
        "metrics": {
          "a003_geometry_is_not_euler": true,
```

```
          "qualified_euler_sources": 0
        },
        "reason": "No independent time-evolved source-native 3D Euler
vorticity AND physically independent pressure/velocity holdout. A003
geometric atlas cannot supply them.",
        "status": "UNRESOLVED"
      }
    },
    "physical_casimir": "UNRESOLVED",
    "physical_euler": "UNRESOLVED",
    "preregistration_sha256":
"d4645ad23017e5be9ec029772ef3a6b70269bd6322a3caf8c812c169dc512473",
    "release": "v0.3.9",
    "schema": "A065-A003-V039-RESULT-1",
    "seed_results": [
      {
        "admissible_sections": 3,
        "field_npz_path":
"outputs/A003_V039_torus_3_1_direct_SYNTHETIC_3D.npz",
        "field_npz_sha256":
"704017e2d0b890e2af5dbadb897f36a99453d14e47e227c21bed275ae412750f",
        "flow_diagnostics": {
          "div_omega_rms": 7.096748311450792e-18,
          "div_u_rms": 2.697307562480913e-18,
          "pressure_poisson_residual_rms": 2.1771919874940538e-15,
          "projection_relative_error": 0.0004113886306385613,
          "steady_euler_residual_relative": 0.6002262946324506
        },
        "gaussian_a_assumed_NOT_INFERRED": 0.55,
        "geometry_input_nodes": 200,
        "grid_box_length": 12.0,
        "grid_n": 48,
        "heldout_5pct_pass": false,
        "polyline_length": 31.897634535371203,
        "sections": [
          {
            "Gamma_cross_section": 1.0842516916485043,
            "Z_axial": 0.497052614947459,
            "Z_full": 0.5688856359870431,
            "a_Gamma_Z_gaussian_assumption": 0.6135345055979666,
            "epsilon_half_circulation_equivalent": 0.5108009817399454,
            "epsilon_slope_equivalent": 0.6135345055979666,
            "fraction": 0.0,
            "heldout_radial_points": 6,
            "heldout_radial_pressure_gradient_rms_relative":
0.17706570413430675,
            "heldout_velocity_rms_relative": 0.04023267345119424,
            "index": 0,
            "model_predictive_gate": "FAIL",
            "nonaxial_vorticity_fraction_rms": 0.35534448680856595,
            "section_admissible": true,
            "signed_flux_ratio": 1.0
          },
          {
            "Gamma_cross_section": 1.0593551781064203,
            "Z_axial": 0.49363455490495933,
            "Z_full": 0.6085335383941007,
            "a_Gamma_Z_gaussian_assumption": 0.6015183534407029,
            "epsilon_half_circulation_equivalent": 0.500796878853043,
            "epsilon_slope_equivalent": 0.6015183534407029,
            "fraction": 0.25,
            "heldout_radial_points": 6,
            "heldout_radial_pressure_gradient_rms_relative":
0.2642986278211246,
            "heldout_velocity_rms_relative": 0.02909255229508105,
            "index": 50,
            "model_predictive_gate": "FAIL",
            "nonaxial_vorticity_fraction_rms": 0.43452606622742984,
            "section_admissible": true,
            "signed_flux_ratio": 0.9904792706389014
          },
          {
            "Gamma_cross_section": 1.2952468097698393,
            "Z_axial": 0.5356462524612039,
            "Z_full": 1.2705154864831725,
            "a_Gamma_Z_gaussian_assumption": 0.7060306610052707,
            "epsilon_half_circulation_equivalent": 0.5878090824386554,
            "epsilon_slope_equivalent": 0.7060306610052707,
```

```
            "fraction": 0.5,
            "heldout_radial_points": 6,
            "heldout_radial_pressure_gradient_rms_relative":
0.7759791753143694,
            "heldout_velocity_rms_relative": 0.10575815056570849,
            "index": 100,
            "model_predictive_gate": "FAIL",
            "nonaxial_vorticity_fraction_rms": 0.7605277257065379,
            "section_admissible": false,
            "signed_flux_ratio": 0.9908756551023905
          },
          {
            "Gamma_cross_section": 1.0593551781064232,
            "Z_axial": 0.4936345549049587,
            "Z_full": 0.6085335383940951,
            "a_Gamma_Z_gaussian_assumption": 0.6015183534407049,
            "epsilon_half_circulation_equivalent": 0.5007968788530447,
            "epsilon_slope_equivalent": 0.6015183534407049,
            "fraction": 0.75,
            "heldout_radial_points": 6,
            "heldout_radial_pressure_gradient_rms_relative":
0.26429862782112407,
            "heldout_velocity_rms_relative": 0.02909255229508224,
            "index": 150,
            "model_predictive_gate": "FAIL",
            "nonaxial_vorticity_fraction_rms": 0.43452606622742235,
            "section_admissible": true,
            "signed_flux_ratio": 0.9904792706388917
          }
        ],
        "seed_id": "torus_3_1_direct",
        "source_class": "SYNTHETIC_GAUSSIAN_FROM_AUTHENTIC_A003_GEOMETRY",
        "source_file": "xyz/torus_3_1_direct_uniform_s_polyline_N1200.xyz",
        "source_sha256":
"16f34a900af097e893a0ad15c3f0b8d0c35d74dbd6edac11d8d60057a98f16e3"
      },
      {
        "admissible_sections": 1,
        "field_npz_path":
"outputs/A003_V039_torus_3_1_hoste_SYNTHETIC_3D.npz",
        "field_npz_sha256":
"6829d9b2a7b79379f863beed58fabc8934de3e095a8d1df6b4ea37015c46ba82",
        "flow_diagnostics": {
          "div_omega_rms": 5.825697732836673e-18,
          "div_u_rms": 1.9846966871649178e-18,
          "pressure_poisson_residual_rms": 9.486865392365979e-16,
          "projection_relative_error": 0.00030763691285507443,
          "steady_euler_residual_relative": 0.7095931860480317
        },
        "gaussian_a_assumed_NOT_INFERRED": 0.55,
        "geometry_input_nodes": 200,
        "grid_box_length": 12.0,
        "grid_n": 48,
        "heldout_5pct_pass": false,
        "polyline_length": 18.399253164416145,
        "sections": [
          {
            "Gamma_cross_section": 0.8068227426465143,
            "Z_axial": 0.49400537105850784,
            "Z_full": 0.6044073668059304,
            "a_Gamma_Z_gaussian_assumption": 0.45795453279590514,
            "epsilon_half_circulation_equivalent": 0.3812721579797999,
            "epsilon_slope_equivalent": 0.45795453279590514,
            "fraction": 0.0,
            "heldout_radial_points": 6,
            "heldout_radial_pressure_gradient_rms_relative":
0.40755828923072307,
            "heldout_velocity_rms_relative": 0.10628165019364283,
            "index": 0,
            "model_predictive_gate": "FAIL",
            "nonaxial_vorticity_fraction_rms": 0.4273892434059213,
            "section_admissible": false,
            "signed_flux_ratio": 0.7223119909511515
          },
          {
            "Gamma_cross_section": -0.23154450019994158,
            "Z_axial": 0.6130528483562308,
            "Z_full": 2.111648742341015,
```

```
            "a_Gamma_Z_gaussian_assumption": 0.1179765191474862,
            "epsilon_half_circulation_equivalent": 0.09822189502457405,
            "epsilon_slope_equivalent": 0.1179765191474862,
            "fraction": 0.25,
            "heldout_radial_points": 6,
            "heldout_radial_pressure_gradient_rms_relative":
0.7068788606145084,
            "heldout_velocity_rms_relative": 2.134208289120361,
            "index": 50,
            "model_predictive_gate": "FAIL",
            "nonaxial_vorticity_fraction_rms": 0.8424253560048235,
            "section_admissible": false,
            "signed_flux_ratio": 0.1596206177345266
          },
          {
            "Gamma_cross_section": 0.4531093632032387,
            "Z_axial": 0.8883233812426954,
            "Z_full": 1.664557391668613,
            "a_Gamma_Z_gaussian_assumption": 0.19179070515131413,
            "epsilon_half_circulation_equivalent": 0.159676235950913,
            "epsilon_slope_equivalent": 0.19179070515131413,
            "fraction": 0.5,
            "heldout_radial_points": 6,
            "heldout_radial_pressure_gradient_rms_relative":
1.535405054400681,
            "heldout_velocity_rms_relative": 0.37392546411539035,
            "index": 100,
            "model_predictive_gate": "FAIL",
            "nonaxial_vorticity_fraction_rms": 0.6828840149782701,
            "section_admissible": false,
            "signed_flux_ratio": 0.2552213658013598
          },
          {
            "Gamma_cross_section": 0.9932503115808042,
            "Z_axial": 0.48721001498788474,
            "Z_full": 0.5727797977635813,
            "a_Gamma_Z_gaussian_assumption": 0.5676892525773476,
            "epsilon_half_circulation_equivalent": 0.4726323049379377,
            "epsilon_slope_equivalent": 0.5676892525773476,
            "fraction": 0.75,
            "heldout_radial_points": 6,
            "heldout_radial_pressure_gradient_rms_relative":
0.6655953951229963,
            "heldout_velocity_rms_relative": 0.01093319913149436,
            "index": 150,
            "model_predictive_gate": "FAIL",
            "nonaxial_vorticity_fraction_rms": 0.3865150142749329,
            "section_admissible": true,
            "signed_flux_ratio": 0.9498552435835919
          }
        ],
        "seed_id": "torus_3_1_hoste",
        "source_class": "SYNTHETIC_GAUSSIAN_FROM_AUTHENTIC_A003_GEOMETRY",
        "source_file": "xyz/torus_3_1_hoste_uniform_s_polyline_N1200.xyz",
        "source_sha256":
"123cd798fc754269ac2baf4225b4c8097c2e47cb5d46b9e814e7f2d74712906d"
      }
    ],
    "source_class": "FOURIER_GEOMETRIC_SEEDS_ONLY_NOT_NATIVE_EULER"
  },
  "v040_volume_euler": {
    "gates": {
      "G37": "PASS",
      "G38": "PASS",
      "G39": "PASS",
      "G40": "PASS",
      "G41": "UNRESOLVED",
      "G42": "PASS"
    },
    "physical_PKLSA": "UNRESOLVED",
    "source_class": "GEOMETRY_SEEDED_GAUSSIAN_EULER_IVP_DIAGNOSTIC_ONLY"
  },
  "v041_arc_gpu": {
    "device": {
      "backend": "backend_type.level_zero",
      "device_name": "Intel(R) Arc(TM) A770 Graphics",
      "dpctl_version": "0.22.1",
      "dpnp_version": "0.20.0",
```

```
      "has_native_fp64": false,
      "sycl_selector": "EXPLICIT_INTEL_ARC_A770_LEVEL_ZERO"
    },
    "gates": {
      "G43": {
        "metrics": {
          "backend": "backend_type.level_zero",
          "device_name": "Intel(R) Arc(TM) A770 Graphics",
          "dpctl_version": "0.22.1",
          "dpnp_version": "0.20.0",
          "has_native_fp64": false,
          "sycl_selector": "EXPLICIT_INTEL_ARC_A770_LEVEL_ZERO"
        },
        "reason": "Explicit Arc A770 Level Zero GPU queue; no fallback",
        "status": "PASS"
      },
      "G44": {
        "metrics": {
          "cpu_seconds": 0.0871023000072455,
          "error": {
            "omega": 2.227848029383319e-07,
            "p": 2.3501380850661573e-07,
            "u": 1.0568855832984928e-07
          },
          "gpu_div_u_rms": 1.1628095153426443e-09,
          "gpu_seconds": 0.5598833999974886,
          "grid": 24
        },
        "reason": "Same operator/initial data GPU FP32 vs independent NumPy
FP64 Taylor-Green (not a PKLSA physical test)",
        "status": "PASS"
      },
      "G45": {
        "metrics": {
          "grids": [
            48,
            64,
            96
          ],
          "seeds": [
            "torus_3_1_direct",
            "torus_3_1_hoste"
          ]
        },
        "reason": "A003 trajectories GPU FP32 vs NumPy FP64 at frozen
48/64/96",
        "status": "PASS"
      },
      "G46": {
        "reason": "Periodically reconstructed Euler pressure GPU FP32 vs CPU
FP64; no independent physical pressure measurement",
        "status": "PASS"
      },
      "G47": {
        "reason": "Full three-resolution dual-backend result/manifest with
explicit A770 identity; source-hash fingerprint required",
        "status": "PASS"
      },
      "G48": {
        "reason": "Gaussian initialization from A003 geometry and dependent
numerical pressure cannot qualify native PKLSA physics",
        "status": "UNRESOLVED"
      }
    },
    "physical_PKLSA": "UNRESOLVED",
    "source_class": "GPU_NUMERICAL_EULER_ASSUMED_GAUSSIAN_NOT_PHYSICAL_PKLSA"
  }
}
```


Source manifest


```
{
  "manifest_sha256":
"5e78b5137f9a58a1e04a24ea84938cc533c7f2887b5169a5931d62b7a4f2a1ce",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
```

```
      "catalog_id": "A046",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "synthetic-euler",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifi
ers\\A065_Triadic_Casimir_Blind\\A065-v0.4.2\\runs\\A065-v0.4.2__20261019T211
624812525Z-14d1834203\\FRAMEWORK_INSTANCE\\A065-v0.4.2\\data\\D01_EULER\\case
s.json",
      "provenance_family": "synthetic-euler",
      "registry_hint": {
        "catalog_id": "A046",
        "hits": []
      },
      "role": "analytic_control",
      "sha256":
"9c4da362ab58a546fe8aff349a024dcc7fb63218ef30053dceee3d6659c574ad",
      "source_id": "D01",
      "version": "v0.4.0"
    },
    {
      "catalog_id": "E010",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "knotplot-relaxed-3_1",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifi
ers\\A065_Triadic_Casimir_Blind\\A065-v0.4.2\\runs\\A065-v0.4.2__20261019T211
624812525Z-14d1834203\\FRAMEWORK_INSTANCE\\A065-v0.4.2\\data\\D01_EULER\\pkls
a\\trefoil_KnotPlot_relaxed_N300.txt",
      "provenance_family": "knotplot-relaxed-3_1",
      "registry_hint": {
        "catalog_id": "E010",
        "hits": []
      },
      "role": "geometry_diagnostic",
      "sha256":
"a60b1a95a00110e3cb32087e7b4328d16e967379ebf97d60117b0ec05692ffb2",
      "source_id": "PKLSA",
      "version": "v0.4.0"
    },
    {
      "catalog_id": "CANON",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "synthetic-link",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifi
ers\\A065_Triadic_Casimir_Blind\\A065-v0.4.2\\runs\\A065-v0.4.2__20261019T211
624812525Z-14d1834203\\FRAMEWORK_INSTANCE\\A065-v0.4.2\\data\\D02_CLOCK_LINK\
\single_surface.json",
      "provenance_family": "synthetic-link",
      "registry_hint": {
        "catalog_id": "CANON",
        "hits": []
      },
      "role": "toy_control",
      "sha256":
"8e4bcfcdf186e539f0cdeafd9cbbf9f31dbe7bbfb95c14d89099e6c31f760af8",
      "source_id": "D02",
      "version": "v0.4.0"
    },
    {
      "catalog_id": "CANON",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "synthetic-sheet",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifi
ers\\A065_Triadic_Casimir_Blind\\A065-v0.4.2\\runs\\A065-v0.4.2__20261019T211
624812525Z-14d1834203\\FRAMEWORK_INSTANCE\\A065-v0.4.2\\data\\D03_TWO_SURFACE
\\conditions.json",
      "provenance_family": "synthetic-sheet",
      "registry_hint": {
        "catalog_id": "CANON",
        "hits": []
      },
      "role": "heldout_control",
      "sha256":
"fff8860c2597840d0f15906a43f8710793c0909ebfdde1b1eb6f64aea7863adc",
      "source_id": "D03",
      "version": "v0.4.0"
```

```
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_f
alsifiers\\A065_Triadic_Casimir_Blind\\A065-v0.4.2\\runs\\A065-v0.4.2__202610
19T211624812525Z-14d1834203\\FRAMEWORK_INSTANCE\\A065-v0.4.2"
}
```


15 Scientific interpretation

Physical Casimir force and gravitation remain unvalidated.

16 Reproduction

Run native FREEZE and FULL via the installation bridge.